OPSINTEL AI

Evidence-first pipeline investigation

Investigate incidents with traceable evidence and reviewer-governed memory

EXATHON INDIA 2026 / TRACK 04: THE WORK BEHIND THE WORK

Public prototype • MIT licensed • Synthetic + public evidence

THE PROBLEM

On-call engineers piece together pipeline incidents from scattered evidence.

MANUAL

Run status, counts, and logs

A failed run reports 80 of 100 expected rows. The engineer must connect run history, counts, and logs before choosing the next check.

OPSINTEL

01 Select the failed run

02 Review cited status, counts, and logs

03 Keep facts separate from candidate explanations

04 Cause stays unconfirmed until review

54% of survey respondents reported a significant outage costing more than $100k (Uptime Institute, 2024). OpsIntel time saving remains unmeasured.

PRODUCT WORKFLOW

Exasol evidence becomes a reviewable investigation

01 **Pipeline runs and logs** →

02 **Exasol SQL analytics** →

03 **FastAPI rules and candidates** →

04 **Dashboard + optional LLM narrative** →

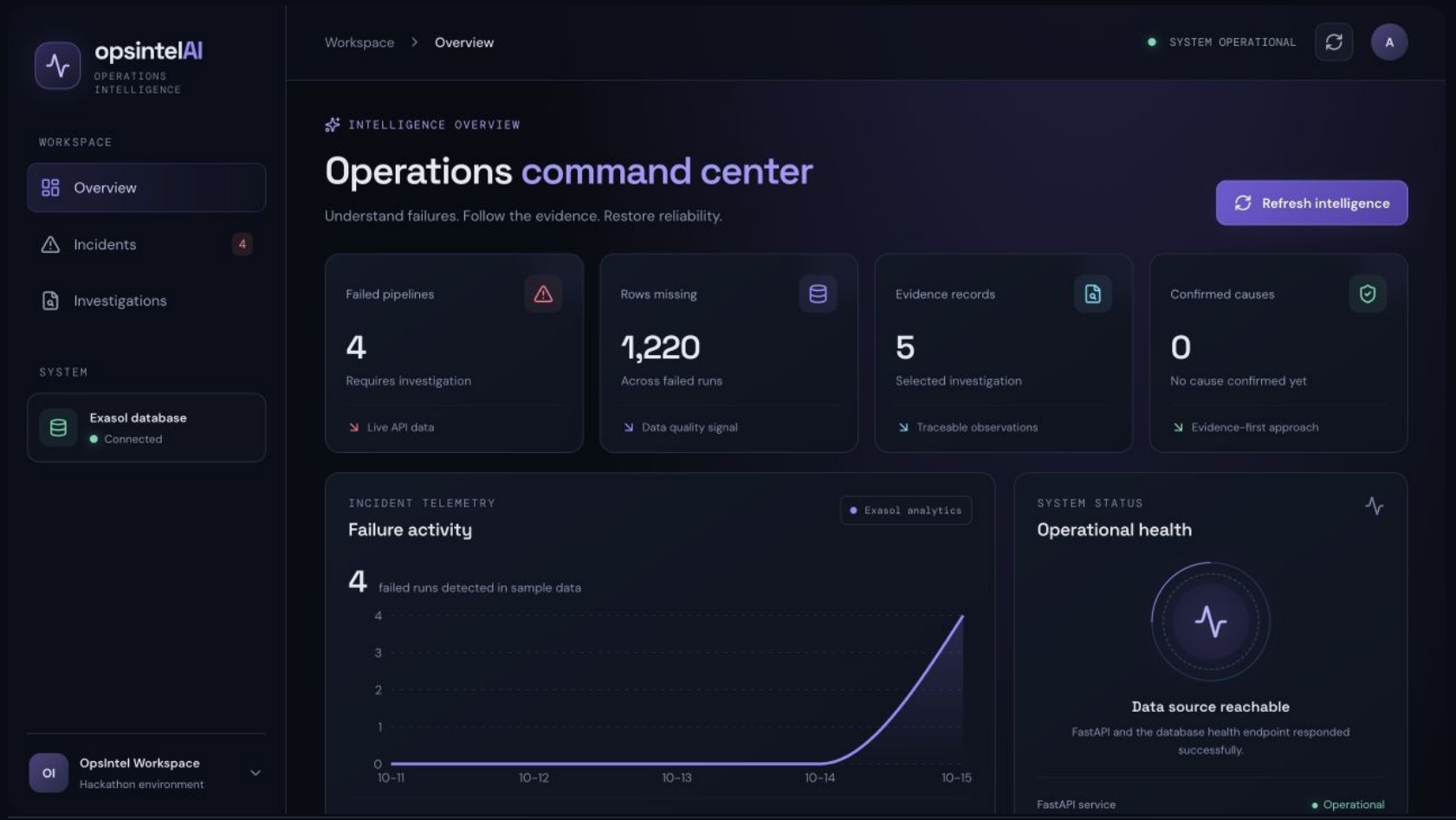
05 **Investigator submits; reviewer decides**

Safety boundary

The deterministic classifier proposes symptoms and candidates. An optional LLM only summarizes retrieved evidence. A separate authorized reviewer checks criteria and evidence before Exasol stores a confirmation. No repair runs automatically.

DASHBOARD EXAMPLE

Exasol dashboard view · synthetic demo telemetry



4

failed pipeline runs

1,220

aggregate missing rows in the captured 4-case baseline

✓ Captured baseline: 4 incidents
✓ Current seed: 11 incident scenarios
✓ All values are synthetic

INVESTIGATION EXPERIENCE

Every recommendation points back to source evidence

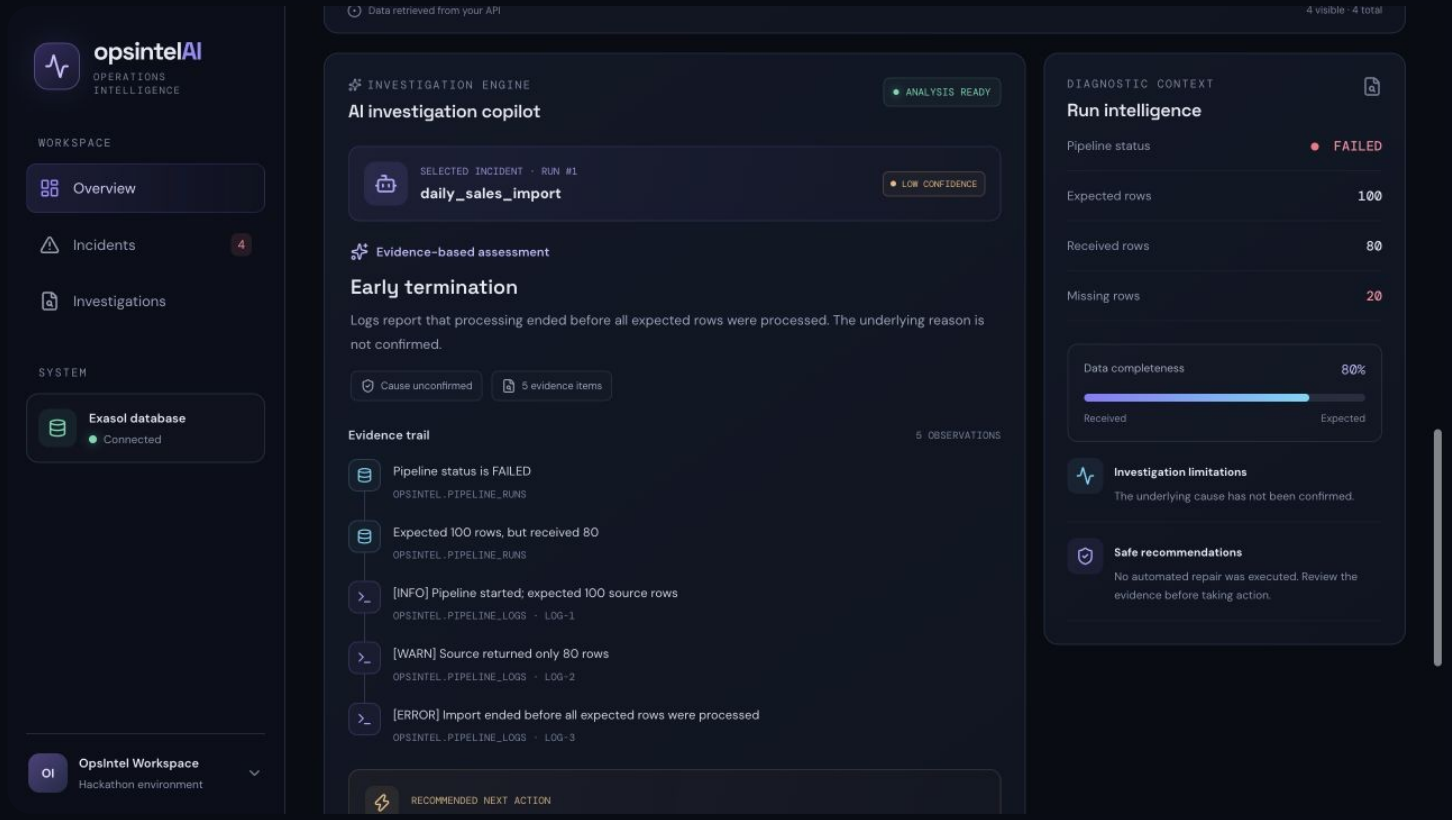
Selected failure vs last success

daily_sales_import

FAILED VS LAST SUCCESS

Failed: 80 / 100 rows (80%)
Last success: 100 / 100 (100%)
Change: -20 rows / -20 pp

EVIDENCE TRAIL
PIPELINE_RUNS: status and row counts
LOG-1: expected 100 source rows
LOG-2: source returned only 80 rows
LOG-3: import ended before completion



CANDIDATES

Source shortfall or early termination
Confidence: low

HUMAN CONFIRMATION

Not yet verified
No automated repair

WHY EXASOL

Exasol computes incident context before FastAPI responds

ANALYTICAL FEATURES

01 Previous run status with LAG

02 Rolling actual-row baseline

03 Rolling failure rate

04 Missing rows and completeness

05 Parallel in-memory SQL execution

LIVE QUERY USED BY /analytics/trend

```
SELECT RUN_DAY, COUNT(*) AS TOTAL_RUNS,
       SUM(IS_INCIDENT) AS INCIDENT_COUNT,
       ROUND(AVG(COMPLETENESS_PCT), 1) AS AVG_COMPLETE
FROM OPSINTEL.PIPELINE_HEALTH_FEATURES
GROUP BY RUN_DAY;
```

VERIFIED EXASOL OUTPUT

12 runs, 11 incidents, 64.1% average completeness

Recorded Exasol Personal output for the synthetic demo schema; it is not a live service check from this run.

TEMPORAL DEPENDENCY CORRELATION

Exasol CONNECT BY surfaces time-correlated downstream failures

01

Window analytics at scale

LAG, rolling baselines and failure rates compare each run with recent history inside Exasol.

02

Failure cascade

CONNECT BY retains downstream failures within a 120-minute window. Temporal correlation only. Causation not confirmed.

03

Operational evidence layer

FastAPI returns the same SQL-derived metrics and source records that judges can inspect in the dashboard.

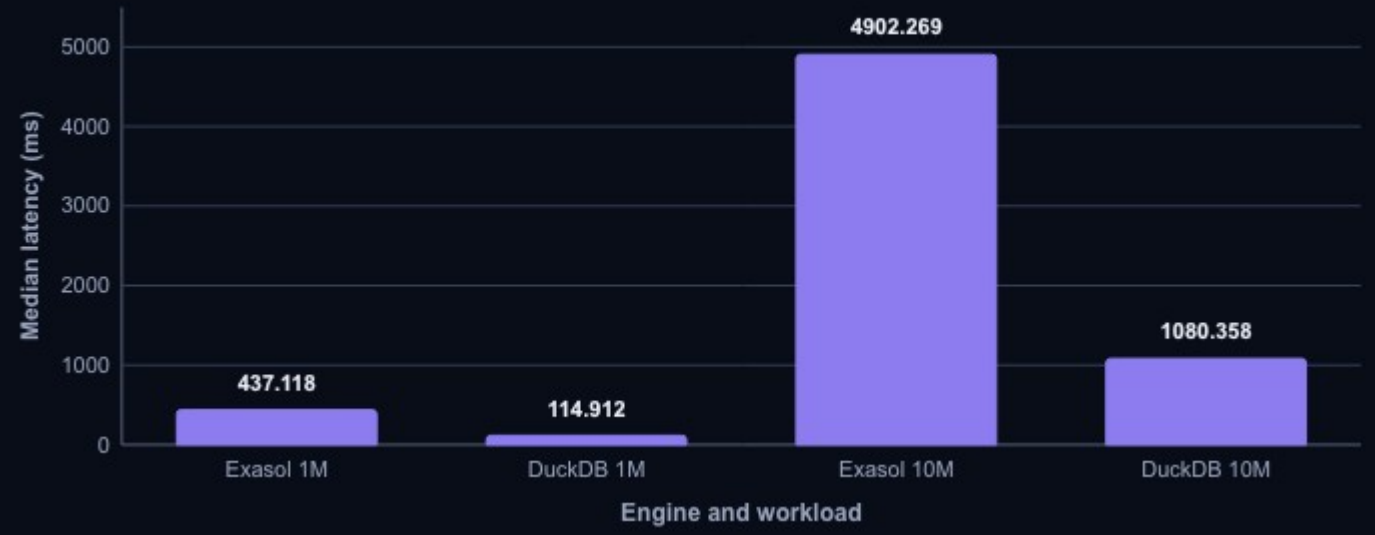
IMPLEMENTED CASCADE QUERY

customer_sync → **daily_reporting** · **customer_orders_sync** · **billing_customer_export**

Recorded result: customer_sync is associated with 3 later failed runs in a 120-minute window. Bulk import loads benchmark history; matching is not an Exasol UDF.

LOCAL SINGLE-USER BENCHMARK

At 10M rows, DuckDB was 4.5× faster



Source figure (non-editable). 10M medians: Exasol 4902.269 ms; DuckDB 1080.358 ms.

200

automated tests passed

7 / 10

project labels
synthetic: 27/27

QUERY CACHE

OFF

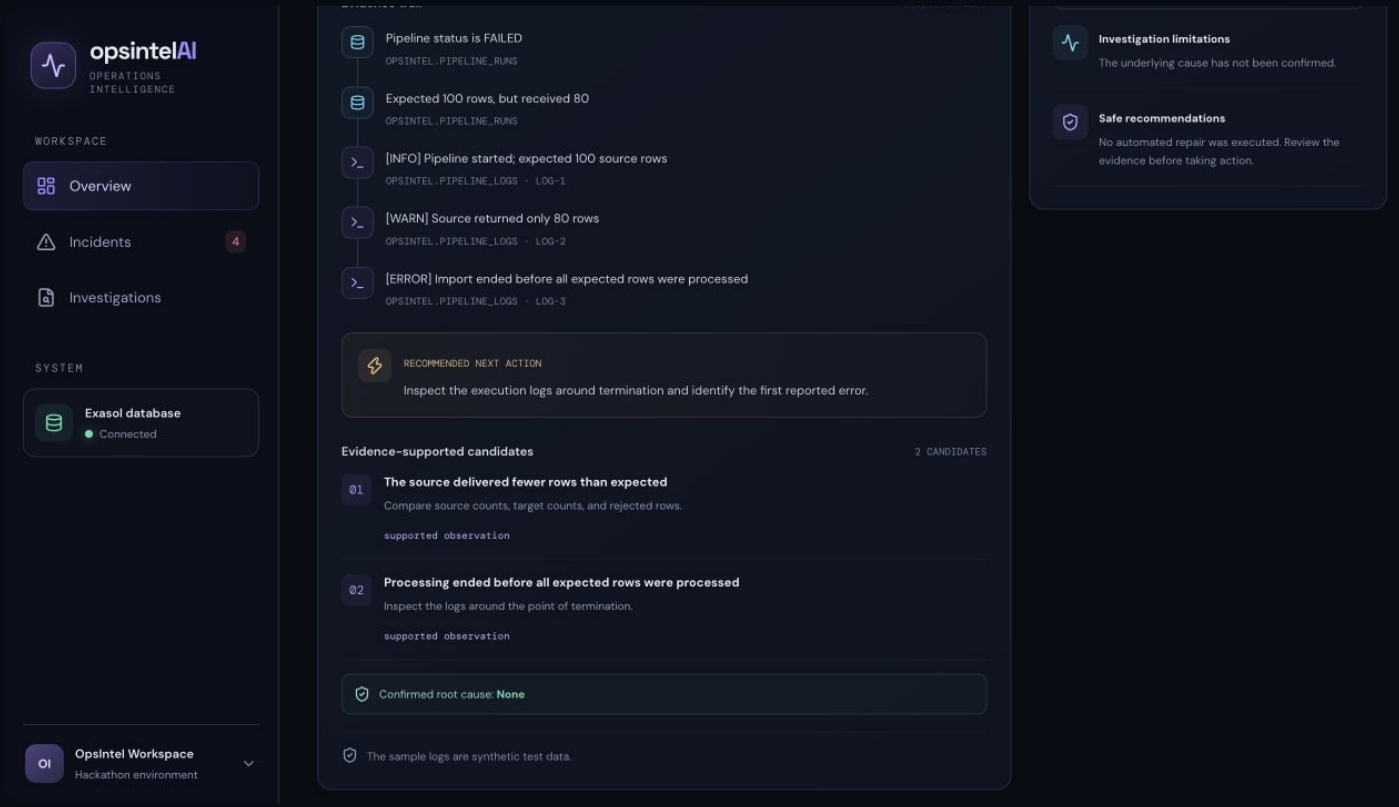
REPETITIONS

7 RUNS

Recorded local benchmark only. Same query, 7 runs per engine, cache off. At 10M rows DuckDB was 4.5× faster in this run. No production or concurrency claim.

TRUST AND SAFETY

Verified memory requires a separate reviewer



OBSERVATION

Source-backed facts

CANDIDATE

Evidence-supported possibility

CONFIRMATION

Authorized reviewer + evidence checklist

Investigator submits cited evidence. A separate reviewer checks criteria and conflicts. Exasol stores approval and audit history; FastAPI returns future matches as candidates only.

VALIDATION ROADMAP

Measured evidence and remaining validation

VERIFIED IMPLEMENTATION

01 Live Exasol memory workflow verified

02 Restart persistence; retrieval score 0.5855

03 Reviewer race guard: 1 HTTP 200 / 1 HTTP 409

04 200 tests; challenge 16/16; empty-history benchmark measured

LIMITS OF CURRENT EVIDENCE

01 Test identities only; test confirmation revoked

02 No active authorized human-confirmed memory

03 Public corpus: 7/10 project labels; no external adjudication

04 Human-impact study: 0 participants

NEXT VALIDATION

01 Obtain authorized human review of source-backed evidence

02 Build independently labelled evaluation set

03 Measure non-empty memory retrieval and scaling

04 Rehearse browser demo and clean-clone setup

Benchmark: empty history only — median 51.627 ms; p95 64.753 ms. No active confirmed memories. E2E used temporary test identities; confirmation was revoked. No human validation, scaling, independent-evaluation or impact result is claimed.

OPSINTEL AI

Investigate with evidence.
Keep causes reviewable.
Confirm only after authorized review.

github.com/Anafa-thabassum/opsintel-ai

Technical workflow verified. Authorized human validation remains pending.